

VTU-ETR Seat No.

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Question Paper Version : A

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) November – 2022

**Computer Science and Engineering /
Information Science and Engineering**



Time: 3 hrs.

Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
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PART – I
(Research Methodology)

[50 MARKS]

1. Research conducted to solve an immediate problem is
 - a) Fundamental Research
 - b) Analytical Research
 - c) Action Research
 - d) Survey
2. Research pertaining to pure mathematics or natural laws is the example of
 - a) Qualitative Research
 - b) Empirical Research
 - c) Analytical Research
 - d) Fundamental Research
3. Research to study the effect of certain Policies, Plans and Program is
 - a) Applied Research
 - b) Descriptive Research
 - c) Evaluation Research
 - d) Causal Research
4. The first step of Research process is
 - a) Research design
 - b) Analysis of data
 - c) Review of literature
 - d) Identification of research problem
5. Which of the following helps to compare two or more variables
 - a) Classification
 - b) Tabulation
 - c) Survey
 - d) Literature review
6. Choose the best answer, which is the most reliable source of information for your literature review :
 - a) A TV documentary
 - b) A Newspaper article
 - c) A peer reviewed research article
 - d) A relevant chapter from a text book
7. Which literature review involves time line based collection of literature review
 - a) Narrative method
 - b) Systematic method
 - c) Meta – analysis method of literature review
 - d) Meta – synthesis method of literature review.
8. Narrative literature review method is also known as
 - a) Advanced method
 - b) Scientific method
 - c) Traditional method
 - d) Systematic method.
9. A review of the literature should sort out the information into a _____ framework
 - a) Conceptual
 - b) Selective
 - c) Theoretical
 - d) Universal
10. The internet should not be used for finding published literature
 - a) True
 - b) False
11. Which one among the following is the most comprehensive source of population data?
 - a) National sample surveys
 - b) Demographic health surveys
 - c) National family health surveys
 - d) Census
12. Which one is called non – probability sampling?
 - a) Cluster sampling
 - b) Systematic sampling
 - c) Stratified sampling
 - d) Quota sampling

13. Which scale is the simplest form of measurement
a) Interval b) Ratio c) Nominal d) Ordinal
14. Absolute zero exists in
a) Ratio scale b) Interval scale c) Nominal scale d) Ordinal scale
15. Which data is cheaper to collect?
a) Primary data b) Collective data c) New data d) Secondary data
16. After the data has been processed and analysed, the research process requires
a) Interpretation of data b) Presentation of data
c) Reporting of data d) Testing of data
17. The process of marking segments of data with symbols, descriptive words or category names is known as
a) Concurring b) Coding c) Coloring d) Segmentation
18. The most repeated value in the series is called as
a) Mean b) Mode c) Median d) Range
19. A study in statistics that helps to interpret the variability of data known as
a) Standard deviation b) The measure of central tendency
c) The measure of dispersion d) None of the above
20. For a frequency distribution of a variable X, mean = 32 and median = 30. The distribution is
a) Positive skewed b) Negatively skewed c) Mesokurtic d) Platykurtic
21. Which among the following is a probability sampling method?
a) Judgemental sampling b) Quota sampling
c) Simple random sampling d) Convenience sampling
22. The difference between a static and the parameters is called as
a) Non - random b) Probability c) Sampling error d) Random
23. Sampling error increases as we increases the sampling size
a) True b) False c) True to some extent d) None of the above
24. Population under research study is referred as
a) Standard Population b) Final Population
c) Infinite Population d) Target Population
25. The process of finding true but unknown value of population parameter is called as
a) Statistical inference b) Estimation
c) Guess d) Biased
26. Arrange the following steps in the process of hypothesis testing in proper sequence
A) Select the level of significance B) Setup null and alternative hypothesis
C) Establish the decision rule D) Performance computations
E) Select test statistics F) Draw conclusion
Choose the correct answer from the options given below :
a) A, B, C, D, E, F b) A, B, E, D, C, F c) B, A, C, D, E, F d) B, A, E, C, D, F

- VER-A : CS/IS 3 of 10**

39. Which of the following is not the part of the exploratory factor analysis process
- Extracting factors
 - Determining the number of factors before the analysis.
 - Rotating the factors
 - Refining and interpreting the factors
40. If a researcher wanted to determine which variable were associated with which factors she/he would look at
- Factor scores
 - Factor loadings
 - Factor
 - Factor associations
41. What is a complementary event for “At least one head appears” if two coins are tossed simultaneously?
- Exactly one head appears
 - Atleast one tail appears
 - Atmost one tail appears
 - None head appears
42. A bag contains 7 Red and 4 Blue Balls. Two Balls are drawn at random with replacement. The probability of getting the Balls of different colors is :
- $28/121$
 - $56/121$
 - $\frac{1}{2}$
 - None of the above
43. If two coins are tossed simultaneously, what is the probability of getting atleast one head?
- $1/2$
 - $1/3$
 - $1/4$
 - $3/4$
44. What are Independent Events?
- If the outcome of one event does not affect the outcome of another
 - If the outcome of one event affects the outcome of another
 - Any one of the outcome of one event does not affect the outcome of another.
 - Any one of the outcome of one event does affect the outcome of another
45. In class, 30% of students study Hindi, 45% study Maths and 15% study Both Hindi and Maths. If a student is randomly selected, what is the probability that he/she study Hindi and Maths?
- $1/5$
 - $3/5$
 - $2/5$
 - $2/7$
46. The normal distribution
- Mean = Variance = 1
 - Standard deviation = Variance = 0
 - Mean = 0 , Variance = 1
 - Mean = Standard deviation
47. The shape of normal distribution curve
- Bell shaped
 - Flat
 - Circular
 - Spilled
48. In a Binomial distribution, if ‘n’ is the number of trials and ‘p’ is the probability success, then the mean value is
- np
 - n
 - p
 - np (1-p)
49. If ‘m’ is the Poisson distribution, the standard deviation is
- $(\sqrt{m})$
 - m^2
 - m
 - m/2
50. The mean of hypergeometric distribution is
- $n * K / N - 1$
 - $n * K - 1 / N$
 - $n - 1 * K / N$
 - $n * K / N$

PART – II
Discipline Oriented Section
(Computer Science and Engineering /
Information Science and Engineering)

[50 MARKS]

51. If $A \cap B = A$ then $A \cup B$ is equal to?
a) A b) ϕ c) A' d) B
52. Which of the following statements is a proposition?
a) Next month I will pass ETR examination b) Oh, It is awesome
c) The sum of $x + 5$ is 7 d) $15 \times 321 = 54321$
53. The range of the relation $R = \{(x, x^2) \mid x \text{ is a prime number less than } 13\}$ is
a) $\{4, 9, 25, 49, 121\}$ b) $\{1, 2, 3, 5, 7\}$
c) $\{2, 3, 5, 7, 11\}$ d) $\{1, 4, 9, 25, 49, 121\}$
54. In how many ways can 3 or more persons can be selected from 12 persons?
a) 2200 b) 220×12 c) 12×3 d) 4017
55. Let G be a simple graph of order n . If the size of G is 56 and size of $\bar{G}$ is 80 then what is n ?
a) 34 b) 17 c) $50 - (80/2) = 10$ d) $80 - 56 = 24$
56. _____ is the process that groups data and procedures into an entity called objects.
a) Object development methodology b) Linear programming
c) Structured programming d) Object oriented system development
57. In the _____ phase the design model is built based on the analysis model.
a) System Design b) Application c) Object design d) Analysis
58. _____ deals with the static process view of a system, from the perspective of a real or prototype case.
a) Component diagram b) Object diagram
c) Deployment diagram d) State diagram
59. _____ denotes the aspirations of the users and the responsibility of the system to its users.
a) UML b) OOA c) Use-cases d) Association
60. The output of object analysis is a description of the _____ and the user requirements.
a) Problem b) Solution c) Quality assurance d) Use cases
61. If a transmission has packet delays 10ms, 12ms and 9ms in a communication, then what this indicates in data transmission?
a) Delay b) Throughput c) Jitter d) Efficiency
62. A noisy channel has transmission speed of 2kbps and a bandwidth of 4kHz. What is the channel Signal to Noise Ration (SNR)
a) $1 + \sqrt{2}$ b) $\sqrt{2} - 1$ c) $1 - \sqrt{2}$ d) None of these

63. A network with CSMA/CD has a bandwidth of 10Mbps. If the maximum propagation time is $40\mu\text{sec}$. What is the maximum frame size?
a) 10 bytes b) 101 bytes c) 100 bytes d) 1 byte
64. Identify the class of IP-address 193.131.80.255.
a) Class C b) Class D c) Class A d) Class E
65. An attacker has captured the login credentials of a Banker by posting fake web page, similar to the Bank web page. This type of attack is known as:
a) Denial of Service (DoS) b) Packet Mistreat
c) Routing Table Poisoning d) Masquerading
66. A page requested by CPU is not available in the main memory is called.
a) Page hit b) Page fault c) Paging d) TLB
67. The content of base and limit registers are updated in.
a) User mode b) Kernel mode c) Unprivileged mode d) Dual mode
68. High level synchronization construct to solve critical section problem is called.
a) Paging b) Swapping c) Monitors d) Testandset
69. Cooperative concurrent reader and writer processes writing to the standard output file leads to.
a) Race condition b) Output varies on the order of execution of processes
c) Output is unpredictable d) All of these
70. Consider the virtual page reference string 1, 2, 3, 2, 4, 1, 3, 2, 4, 1 on a demand paged virtual memory system running on a computer system that has main memory size of 3 page frames which are initially empty. Let LRU, FIFO and OPTIMAL denote the number of page faults under the corresponding page replacement policy. Then,
a) $\text{OPTIMAL} < \text{LRU} < \text{FIFO}$ b) $\text{OPTIMAL} < \text{FIFO} < \text{LRU}$
c) $\text{OPTIML} = \text{LRU} = \text{FIFO}$ d) $\text{OPTIMAL} = \text{FIFO}$
71. What is the minimum time required to compute power (x, n) is x^n in the options given below?
a) $O(\log n)$ b) $O(n)$ c) $O(\log \log n)$ d) $O(n \log n)$
72. n strings each of length n are given. What is time taken to sort the strings in lexicographic order using 2 way merging?
a) $O(n \log n)$ b) $O(n^2 \log n)$ c) $O(n \log \log n)$ d) $O(n \log^2 n)$
73. Number of spanning tree for complete graph with six nodes is
a) 1296 b) 1398 c) 1354 d) 1366
74. Number of ways of parenthesizing 6 matrices is
a) 36 b) 42 c) 64 d) 78

75. For a job assignment problem

J ₁	J ₂	J ₃	J ₄	
10	2	7	8	P ₁
6	4	3	7	P ₂
5	8	1	8	P ₃
7	6	10	4	P ₄

Total cost of optimal (least cost) assignment is

- a) 10 b) 18 c) 13 d) 14

76. In software engineering the most popular term COTS stands for

- a) Citation On The System b) Commercial On The System
c) Commercial Off The Shelf System d) Central Operating Trading System

77. Form based requirement specification in software engineering is classified under

- a) New Language Specification b) Structured Natural Language
c) Graphical Notations d) Mathematical Specifications

78. Sequence diagrams used in software development are classified under

- a) Static model b) Dynamic model
c) Probabilistic model d) Stochastic model

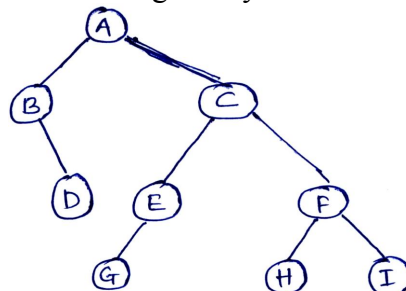
79. In the following software testing method individual program units or object classes are tested

- a) Unit testing b) Component testing c) System testing d) All of these

80. Software pricing depends upon

- a) Travel cost b) Hardware and software cost
c) Training cost d) All of these

81. What is the post order traversal of following binary tree?



- a) A B D C G E H I F b) D B G E H I F C A
c) D B G E F H I C A d) A B C D E F G H I

82. Insertion of an element at the middle of a singly linked list requires the modifications of how many pointers?

- a) 2 b) 1 c) 3 d) 4

83. Which type of traversal of binary search tree outputs the value in sorted order?

- a) Pre order b) In order c) Post order d) None

84. The searching operation in an array is done using.

- a) Key and Index b) Only key c) Only Index d) None of these

85. A binary search tree is generated by inserting in order the following integers (left to right).
50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24. The number of nodes in the left subtree and right subtree of the root respectively is.
a) (4, 7) b) (8, 3) c) (7, 4) d) (3, 8)
86. Consider the following processor design characteristics:
I. Register – to – Register arithmetic operation only.
II. Fixed-length instruction format.
III. Hardwired control unit.
Which of the characteristics above are used in the design of a RISC processor?
a) I and II only b) II and III only c) I and III only d) I, II and III
87. A processor has 40 distinct instruction and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is _____.
a) 16 bits b) 17-bits c) 18-bits d) 19-bits
88. Consider the following code sequence having five instructions I₁ to I₅, each of these instruction has the following format:
OP R_i, R_j, R_k
Where operation OP is performed on contents of register R_j and R_k and the result is stored in register R_i.
I₁ : ADD R₁, R₂, R₃
I₂ : MUL R₇, R₁, R₃
I₃ : SUB R₄, R₁, R₅
I₄ : ADD R₃, R₂, R₄
I₅ : MUL R₇, R₈, R₉
Consider the following three statements:
S₁ : There is an anti-dependence between instruction I₂ and I₅
S₂ : There is an anti-dependence between instruction I₂ and I₄
S₃ : Within an instruction pipeline an anti-dependence always creates one or more stalls.
Which one of above statements is/are correct?
a) Only S₁ is true b) Only S₂ is true
c) Only S₁ and S₃ are true d) Only S₂ and S₃ are true
89. Consider the following sequence of Micro-Operations
MBR ← PC
MAR ← X
PC ← Y
Memory ← MBR
Which one of the following is a possible operation performed by this sequence?
a) Instruction fetch b) Operand fetch
c) Conditional fetch d) Initiation of interrupt service
90. Register renaming is done in pipelined processor
a) as an alternative to register allocation at compile time
b) for efficient access to function parameter and local variable
c) to handle certain kinds of hazards
d) as part of address translation

91. The relational operator <OP> described below represents the following:

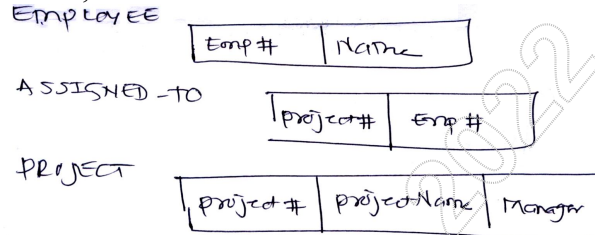
$$R = P \text{ <OP> } Q$$

$$R = \{t \mid t \in p \wedge t \notin \theta\} \text{ and}$$

$$0 \leq |R| \leq |P|$$

- a) The difference '−' operator
b) The projection 'π' operator
c) The division '÷' operator
d) None of the above

92. Consider the schema as below,



The query

$$\text{EMPLOYEE} \bowtie (\text{ASSIGNED - TO} \div \pi_{\text{project\#}}(X))$$

X = $\sigma_{(\text{project\#} = 'P1' \vee \text{project\#} = 'P2')}$ (ASSIGNED-TO) will results in to:

- a) Gathering details at all employees.
b) Gathering details at all employees working on project 'P₁' or project 'P₂'.
c) Gathering details at all employees working on both project 'P₁' and 'P₂'.
d) None of these.

93. Consider the following schema employees:

Emp#	Name	DOB	Sex
------	------	-----	-----

The SQL provision to ensure the attribute 'sex' takes the values either 'M' or 'F' is

- a) ensure (sex in ('M', 'F'))
b) insist (sex in ('M', 'F'))
c) check (sex in ('M', 'F'))
d) Need to write stored procedure.

94. Stored procedures are used to:

- a) enforce business rules and improve performance
b) to make code easier to maintain.
c) code will be stored once for use by multiple applications
d) All of these

95. To set transaction in read only access mode is through:

- a) Setting isolation level in read uncommitted
b) Setting isolation level in read committed
c) Setting isolation level in serializable
d) Setting isolation level in repeatable read mode.

96. Customers arrive at a reception counter at an average interval of 10 minutes and the receptionist takes an average of 6 minutes for one customer. Determine the average queue length.

- a) 9/10
b) 7/10
c) 11/10
d) 3/10

97. Which of the following statements are not true of simulation?

- a) Simulation model cannot prescribe what should be done about a problem.
b) Simulation models can be used to study alternative solutions to a problem.
c) Simulation models the behavior of a system.
d) The equations describing the operating characteristics of the system are known.

98. Simulation models can be used to obtain operating characteristics in less time than with real system using a feature of simulation called.
- a) Microseconds
 - b) Warp seed
 - c) Time compression
 - d) None of the above
99. A multiple-choice test has 20 questions. There are four choices for each question. A student who has not studied for the test decides to answer all the questions randomly by guessing the answer for each question. Which of the following probability distributions can be used to calculate the students chance of getting at least 20 questions right?
- a) Binomial Distribution
 - b) Poisson Distribution
 - c) Exponential Distribution
 - d) Normal Distribution
100. Which of the following are techniques to perform verification of simulation model?
- a) By using programming skills to write and debug the program in sub-programs.
 - b) By tracing the intermediate results and comparing them with observed outcomes.
 - c) By checking the simulation model output using various input combinations.
 - d) All of these.

VTU-ETR Seat No.

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Question Paper Version : B

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- VER-B : CS/IS 1 of 10**

12. In ANOVA procedure 'Factor' refers to
 a) The dependent variable
 b) The independent variable
 c) Different levels of a treatment
 d) The critical value of F
13. In the ANOVA 'treatment' refers to
 a) Experimental units
 b) Different levels of a factor
 c) a factor
 d) Applying antibiotic to a wound
14. When conducting an ANOVA, F DATA will always fall within what range?
 a) Between 0 and infinity
 b) Between 0 and 1
 c) Between negative infinity and infinity
 d) Between 1 and infinity
15. In one way ANOVA which of the following is used within F – ratio as a measurement of the variance F individual observations
 a) SSE
 b) MSE
 c) MSRT
 d) None of the above
16. In regression analysis, the variable that is being predicted is
 a) The independent variable
 b) The dependent variable
 c) Usually denoted by x
 d) Usually denoted by r
17. If the regression equation is equal to $Y = 23.6 - 54.2x$, then 23.6 is the ____ while $- 54.2$ is the ____ of the regression line.
 a) Slope, intercept
 b) Slope, regression coefficient
 c) Intercept, slope
 d) Radius, intercept
18. How many coefficients do you need to estimate in a simple linear regression model (one independent variable).
 a) 1
 b) 2
 c) 3
 d) 4
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33. Research to study the effect of certain Policies, Plans and Program is
a) Applied Research b) Descriptive Research
c) Evaluation Research d) Causal Research
34. The first step of Research process is
a) Research design b) Analysis of data
c) Review of literature d) Identification of research problem
35. Which of the following helps to compare two or more variables
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a) A TV documentary b) A Newspaper article
c) A peer reviewed research article d) A relevant chapter from a text book
37. Which literature review involves time line based collection of literature review
a) Narrative method b) Systematic method
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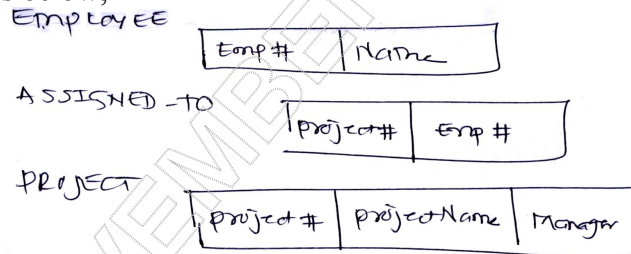
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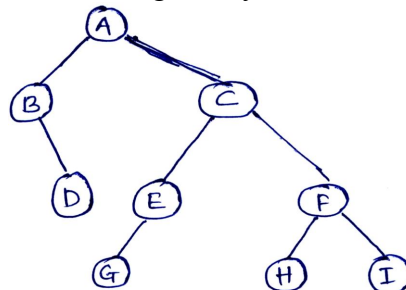
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 d) The equations describing the operating characteristics of the system are known.
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61. What is the post order traversal of following binary tree?



- a) A B D C G E H I F b) D B G E H I F C A
 c) D B G E F H I C A d) A B C D E F G H I
62. Insertion of an element at the middle of a singly linked list requires the modifications of how many pointers?
 a) 2 b) 1 c) 3 d) 4
63. Which type of traversal of binary search tree outputs the value in sorted order?
 a) Pre order b) In order c) Post order d) None
64. The searching operation in an array is done using.
 a) Key and Index b) Only key c) Only Index d) None of these

65. A binary search tree is generated by inserting in order the following integers (left to right).
50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24. The number of nodes in the left subtree and right subtree of the root respectively is.
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I. Register – to – Register arithmetic operation only.
II. Fixed-length instruction format.
III. Hardwired control unit.
Which of the characteristics above are used in the design of a RISC processor?
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OP R_i, R_j, R_k
Where operation OP is performed on contents of register R_j and R_k and the result is stored in register R_i.
I₁ : ADD R₁, R₂, R₃
I₂ : MUL R₇, R₁, R₃
I₃ : SUB R₄, R₁, R₅
I₄ : ADD R₃, R₂, R₄
I₅ : MUL R₇, R₈, R₉
Consider the following three statements:
S₁ : There is an anti-dependence between instruction I₂ and I₅
S₂ : There is an anti-dependence between instruction I₂ and I₄
S₃ : Within an instruction pipeline an anti-dependence always creates one or more stalls.
Which one of above statements is/are correct?
a) Only S₁ is true b) Only S₂ is true
c) Only S₁ and S₃ are true d) Only S₂ and S₃ are true
69. Consider the following sequence of Micro-Operations
MBR ← PC
MAR ← X
PC ← Y
Memory ← MBR
Which one of the following is a possible operation performed by this sequence?
a) Instruction fetch b) Operand fetch
c) Conditional fetch d) Initiation of interrupt service
70. Register renaming is done in pipelined processor
a) as an alternative to register allocation at compile time
b) for efficient access to function parameter and local variable
c) to handle certain kinds of hazards
d) as part of address translation

71. If a transmission has packet delays 10ms, 12ms and 9ms in a communication, then what this indicates in data transmission?
 a) Delay b) Throughput c) Jitter d) Efficiency
72. A noisy channel has transmission speed of 2kbps and a bandwidth of 4kHz. What is the channel Signal to Noise Ratio (SNR)
 a) $1 + \sqrt{2}$ b) $\sqrt{2} - 1$ c) $1 - \sqrt{2}$ d) None of these
73. A network with CSMA/CD has a bandwidth of 10Mbps. If the maximum propagation time is 40μsec. What is the maximum frame size?
 a) 10 bytes b) 101 bytes c) 100 bytes d) 1 byte
74. Identify the class of IP-address 193.131.80.255.
 a) Class C b) Class D c) Class A d) Class E
75. An attacker has captured the login credentials of a Banker by posting fake web page, similar to the Bank web page. This type of attack is known as:
 a) Denial of Service (DoS) b) Packet Mistreat
 c) Routing Table Poisoning d) Masquerading
76. A page requested by CPU is not available in the main memory is called.
 a) Page hit b) Page fault c) Paging d) TLB
77. The content of base and limit registers are updated in.
 a) User mode b) Kernel mode c) Unprivileged mode d) Dual mode
78. High level synchronization construct to solve critical section problem is called.
 a) Paging b) Swapping c) Monitors d) Testandset
79. Cooperative concurrent reader and writer processes writing to the standard output file leads to.
 a) Race condition b) Output varies on the order of execution of processes
 c) Output is unpredictable d) All of these
80. Consider the virtual page reference string 1, 2, 3, 2, 4, 1, 3, 2, 4, 1 on a demand paged virtual memory system running on a computer system that has main memory size of 3 page frames which are initially empty. Let LRU, FIFO and OPTIMAL denote the number of page faults under the corresponding page replacement policy. Then,
 a) OPTIMAL < LRU < FIFO b) OPTIMAL < FIFO < LRU
 c) OPTIML = LRU = FIFO d) OPTIMAL = FIFO
81. If $A \cap B = A$ then $A \cup B$ is equal to?
 a) A b) ϕ c) A' d) B
82. Which of the following statements is a proposition?
 a) Next month I will pass ETR examination b) Oh, It is awesome
 c) The sum of x + 5 is 7 d) $15 \times 321 = 54321$

83. The range of the relation $R = \{(x, x^2) \mid x \text{ is a prime number less than } 13\}$ is
 a) $\{4, 9, 25, 49, 121\}$ b) $\{1, 2, 3, 5, 7\}$
 c) $\{2, 3, 5, 7, 11\}$ d) $\{1, 4, 9, 25, 49, 121\}$
84. In how many ways can 3 or more persons can be selected from 12 persons?
 a) 2200 b) 220×12 c) 12×3 d) 4017
85. Let G be a simple graph of order η . If the size of G is 56 and size of $\bar{G}$ is 80 then what is η ?
 a) 34 b) 17 c) $50 - (80/2) = 10$ d) $80 - 56 = 24$
86. _____ is the process that groups data and procedures into an entity called objects.
 a) Object development methodology b) Linear programming
 c) Structured programming d) Object oriented system development
87. In the _____ phase the design model is built based on the analysis model.
 a) System Design b) Application c) Object design d) Analysis
88. _____ deals with the static process view of a system, from the perspective of a real or prototype case.
 a) Component diagram b) Object diagram
 c) Deployment diagram d) State diagram
89. _____ denotes the aspirations of the users and the responsibility of the system to its users.
 a) UML b) OOA c) Use-cases d) Association
90. The output of object analysis is a description of the _____ and the user requirements.
 a) Problem b) Solution c) Quality assurance d) Use cases
91. What is the minimum time required to compute power (x, n) is x^n in the options given below?
 a) $O(\log n)$ b) $O(n)$ c) $O(\log \log n)$ d) $O(n \log n)$
92. n strings each of length n are given. What is time taken to sort the strings in lexicographic order using 2 way merging?
 a) $O(n \log n)$ b) $O(n^2 \log n)$ c) $O(n \log \log n)$ d) $O(n \log^2 n)$
93. Number of spanning tree for complete graph with six nodes is
 a) 1296 b) 1398 c) 1354 d) 1366
94. Number of ways of parenthesizing 6 matrices is
 a) 36 b) 42 c) 64 d) 78
95. For a job assignment problem

J_1	J_2	J_3	J_4	
10	2	7	8	P_1
6	4	3	7	P_2
5	8	1	8	P_3
7	6	10	4	P_4

Total cost of optimal (least cost) assignment is

- a) 10 b) 18 c) 13 d) 14

96. In software engineering the most popular term COTS stands for
a) Citation On The System b) Commercials On The System
c) Commercial Off The Shelf System d) Central Operating Trading System
97. Form based requirement specification in software engineering is classified under
a) New Language Specification b) Structured Natural Language
c) Graphical Notations d) Mathematical Specifications
98. Sequence diagrams used in software development are classified under
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c) Probabilistic model d) Stochastic model
99. In the following software testing method individual program units or object classes are tested
a) Unit testing b) Component testing c) System testing d) All of these
100. Software pricing depends upon
a) Travel cost b) Hardware and software cost
c) Training cost d) All of these

VTU-ETR Seat No.

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Question Paper Version : C

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) November – 2022

**Computer Science and Engineering /
Information Science and Engineering**



Time: 3 hrs.

Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
8. Candidates should use black ball point pen to shade the checkbox bubble so that it is fully dark and the computer can detect without fail. Shading should be limited to the checkbox bubble only.
9. Shading of multiple checkbox bubbles shall be treated as invalid answer.
10. Partial or incomplete shading/darkening of the bubble be avoided.
11. Use of whiteners/correction fluid to change the shaded/filled checkbox bubble to fill another is prohibited.
12. Erasing of filled checkbox bubble/s to fill another is prohibited.
13. Any type of writing, scribbling etc., on the question paper and ticking the response of a question/s are prohibited. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
14. Only non-programmable scientific calculator is permitted. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
15. Folding/crushing/mutilating of OMR sheet is not permitted.
16. Handover the OMR sheet to the Room Invigilator while leaving the examination hall.

PART – I
(Research Methodology)

[50 MARKS]

1. Which among the following is a probability sampling method?
a) Judgemental sampling b) Quota sampling
c) Simple random sampling d) Convenience sampling
2. The difference between a static and the parameters is called as
a) Non - random b) Probability c) Sampling error d) Random
3. Sampling error increases as we increases the sampling size
a) True b) False c) True to some extent d) None of the above
4. Population under research study is referred as
a) Standard Population b) Final Population
c) Infinite Population d) Target Population
5. The process of finding true but unknown value of population parameter is called as
a) Statistical inference b) Estimation
c) Guess d) Biased
6. Arrange the following steps in the process of hypothesis testing in proper sequence
A) Select the level of significance B) Setup null and alternative hypothesis
C) Establish the decision rule D) Performance computations
E) Select test statistics F) Draw conclusion
Choose the correct answer from the options given below :
a) A, B, C, D, E, F b) A, B, E, D, C, F c) B, A, C, D, E, F d) B, A, E, C, D, F
7. Which among the following is non parametric statistics?
a) t - test b) F - test
c) 2 - test d) Spearman's rank order correlation
8. Which test is normally applied for qualitative data :
a) 't' test b) 'F' test c) X^2 Chi – square test d) 'z' test
9. An investigator commits Type – 1 error in testing hypothesis when he / she
a) Accepts null hypothesis when it is false b) Rejects null hypothesis when it is true
c) Accepts null hypothesis when it is true d) Rejects null hypothesis when it is false
10. If the critical region is located equally in both sides of the sampling distribution of test statistic, the test is called
a) One tailed b) Two tailed c) Right tailed d) Left tailed
11. What is a complementary event for “At least one head appears” if two coins are tossed simultaneously?
a) Exactly one head appears b) Atleast one tail appears
c) Atmost one tail appears d) None head appears

12. A bag contains 7 Red and 4 Blue Balls. Two Balls are drawn at random with replacement. The probability of getting the Balls of different colors is :
 a) $28/121$ b) $56/121$ c) $\frac{1}{2}$ d) None of the above
13. If two coins are tossed simultaneously, what is the probability of getting atleast one head?
 a) $1/2$ b) $1/3$ c) $1/4$ d) $3/4$
14. What are Independent Events?
 a) If the outcome of one event does not affect the outcome of another
 b) If the outcome of one event affects the outcome of another
 c) Any one of the outcome of one event does not affect the outcome of another.
 d) Any one of the outcome of one event does affect the outcome of another
15. In class, 30% of students study Hindi, 45% study Maths and 15% study Both Hindi and Maths. If a student is randomly selected, what is the probability that he/she study Hindi and Maths?
 a) $1/5$ b) $3/5$ c) $2/5$ d) $2/7$
16. The normal distribution
 a) Mean = Variance = 1 b) Standard deviation = Variance = 0
 c) Mean = 0 , Variance = 1 d) Mean = Standard deviation
17. The shape of normal distribution curve
 a) Bell shaped b) Flat c) Circular d) Spilled
18. In a Binomial distribution, if 'n' is the number of trials and 'p' is the probability success, then the mean value is
 a) np b) n c) p d) np (1-p)
19. If 'm' is the Poisson distribution, the standard deviation is
 a) $(\sqrt{m})$ b) m^2 c) m d) m/2
20. The mean of hypergeometric distribution is
 a) $n * K / N - 1$ b) $n * K - 1 / N$ c) $n - 1 * K / N$ d) $n * K / N$
21. What is the function of a post test in ANOVA
 a) Describe those groups that have reliable difference between group means
 b) Set the critical value for the 'F' test (or Chi – Square)
 c) Determine if any statistically significant group differences have occurred
 d) None of these
22. In ANOVA procedure 'Factor' refers to
 a) The dependent variable b) The independent variable
 c) Different levels of a treatment d) The critical value of F
23. In the ANOVA 'treatment' refers to
 a) Experimental units b) Different levels of a factor
 c) a factor d) Applying antibiotic to a wound
24. When conducting an ANOVA , F DATA will always fall within what range?
 a) Between 0 and infinity b) Between 0 and 1
 c) Between negative infinity and infinity d) Between 1 and infinity

25. In one way ANOVA which of the following is used within F – ratio as a measurement of the variance F individual observations
 a) SSE b) MSE c) MSRT d) None of the above
26. In regression analysis, the variable that is being predicted is
 a) The independent variable b) The dependent variable
 c) Usually denoted by x d) Usually denoted by r
27. If the regression equation is equal to $Y = 23.6 - 54.2x$, then 23.6 is the ____ while – 54.2 is the ____ of the regression line.
 a) Slope, intercept b) Slope, regression coefficient
 c) Intercept , slope d) Radius, intercept
28. How many coefficients do you need to estimate in a simple linear regression model (one independent variable).
 a) 1 b) 2 c) 3 d) 4
29. Which of the following is not the part of the exploratory factor analysis process
 a) Extracting factors
 b) Determining the number of factors before the analysis.
 c) Rotating the factors d) Refining and interpreting the factors
30. If a researcher wanted to determine which variable were associated with which factors she/he would look at
 a) Factor scores b) Factor loadings c) Factor d) Factor associations
31. Which one among the following is the most comprehensive source of population data?
 a) National sample surveys b) Demographic health surveys
 c) National family health surveys d) Census
32. Which one is called non – probability sampling?
 a) Cluster sampling b) Systematic sampling
 c) Stratified sampling d) Quota sampling
33. Which scale is the simplest form of measurement
 a) Interval b) Ratio c) Nominal d) Ordinal
34. Absolute zero exists in
 a) Ratio scale b) Interval scale c) Nominal scale d) Ordinal scale
35. Which data is cheaper to collect?
 a) Primary data b) Collective data c) New data d) Secondary data
36. After the data has been processed and analysed, the research process requires
 a) Interpretation of data b) Presentation of data
 c) Reporting of data d) Testing of data
37. The process of marking segments of data with symbols, descriptive words or category names is known as
 a) Concurring b) Coding c) Coloring d) Segmentation

38. The most repeated value in the series is called as
 a) Mean b) Mode c) Median d) Range
39. A study in statistics that helps to interpret the variability of data known as
 a) Standard deviation b) The measure of central tendency
 c) The measure of dispersion d) None of the above
40. For a frequency distribution of a variable X, mean = 32 and median = 30. The distribution is
 a) Positive skewed b) Negatively skewed c) Mesokurtic d) Platykurtic
41. Research conducted to solve an immediate problem is
 a) Fundamental Research b) Analytical Research
 c) Action Research d) Survey
42. Research pertaining to pure mathematics or natural laws is the example of
 a) Qualitative Research b) Empirical Research
 c) Analytical Research d) Fundamental Research
43. Research to study the effect of certain Policies, Plans and Program is
 a) Applied Research b) Descriptive Research
 c) Evaluation Research d) Causal Research
44. The first step of Research process is
 a) Research design b) Analysis of data
 c) Review of literature d) Identification of research problem
45. Which of the following helps to compare two or more variables
 a) Classification b) Tabulation c) Survey d) Literature review
46. Choose the best answer, which is the most reliable source of information for your literature review :
 a) A TV documentary b) A Newspaper article
 c) A peer reviewed research article d) A relevant chapter from a text book
47. Which literature review involves time line based collection of literature review
 a) Narrative method b) Systematic method
 c) Meta – analysis method of literature review
 d) Meta – synthesis method of literature review.
48. Narrative literature review method is also known as
 a) Advanced method b) Scientific method
 c) Traditional method d) Systematic method.
49. A review of the literature should sort out the information into a _____ framework
 a) Conceptual b) Selective c) Theoretical d) Universal
50. The internet should not be used for finding published literature
 a) True b) False

PART – II
Discipline Oriented Section
(Computer Science and Engineering /
Information Science and Engineering)

[50 MARKS]

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- Total cost of optimal (least cost) assignment is
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60. Software pricing depends upon
a) Travel cost b) Hardware and software cost
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61. The relational operator <OP> described below represents the following:

$$R = P \langle OP \rangle Q$$

$$R = \{t \mid t \in p \wedge t \notin \theta\} \text{ and}$$

$$0 \leq |R| \leq |P|$$

- a) The difference '____' operator
b) The projection ' π ' operator
c) The division ' $\div$ ' operator
d) None of the above

62. Consider the schema as below,

EMPLOYEE

Emp #	Name
-------	------

ASSIGNED-TO

Project #	Emp #
-----------	-------

PROJECT

Project #	Project Name	Manager
-----------	--------------	---------

The query

$$\text{EMPLOYEE} \bowtie (\text{ASSIGNED-TO} \div \pi_{\text{project\#}}(X))$$

$X = \sigma_{(\text{project\#} = 'P_1' \vee \text{project\#} = 'P_2')}$ (ASSIGNED-TO) will results in to:

- a) Gathering details at all employees.
b) Gathering details at all employees working on project 'P₁' or project 'P₂'.
c) Gathering details at all employees working on both project 'P₁' and 'P₂'.
d) None of these.

63. Consider the following schema employees:

Emp#	Name	DOB	Sex
------	------	-----	-----

The SQL provision to ensure the attribute 'sex' takes the values either 'M' or 'F' is

- a) ensure (sex in ('M', 'F'))
b) insist (sex in ('M', 'F'))
c) check (sex in ('M', 'F'))
d) Need to write stored procedure.

64. Stored procedures are used to:

- a) enforce business rules and improve performance
b) to make code easier to maintain.
c) code will be stored once for use by multiple applications
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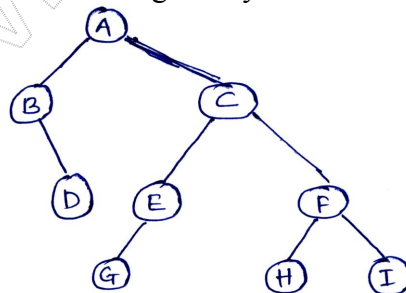
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 - Simulation models can be used to study alternative solutions to a problem.
 - Simulation models the behavior of a system.
 - The equations describing the operating characteristics of the system are known.
68. Simulation models can be used to obtain operating characteristics in less time than with real system using a feature of simulation called.
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- By using programming skills to write and debug the program in sub-programs.
 - By tracing the intermediate results and comparing them with observed outcomes.
 - By checking the simulation model output using various input combinations.
 - All of these.

71. What is the post order traversal of following binary tree?



- A B D C G E H I F
 - D B G E F H I C A
 - D B G E H I F C A
 - A B C D E F G H I
72. Insertion of an element at the middle of a singly linked list requires the modifications of how many pointers?
- 2
 - 1
 - 3
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73. Which type of traversal of binary search tree outputs the value in sorted order?
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 I. Register – to – Register arithmetic operation only.
 II. Fixed-length instruction format.
 III. Hardwired control unit.
 Which of the characteristics above are used in the design of a RISC processor?
 a) I and II only b) II and III only c) I and III only d) I, II and III
77. A processor has 40 distinct instruction and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is ____
 a) 16 bits b) 17-bits c) 18-bits d) 19-bits
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 Where operation OP is performed on contents of register R_j and R_k and the result is stored in register R_i.
 I₁ : ADD R₁, R₂, R₃
 I₂ : MUL R₇, R₁, R₃
 I₃ : SUB R₄, R₁, R₅
 I₄ : ADD R₃, R₂, R₄
 I₅ : MUL R₇, R₈, R₉
 Consider the following three statements:
 S₁ : There is an anti-dependence between instruction I₂ and I₅
 S₂ : There is an anti-dependence between instruction I₂ and I₄
 S₃ : Within an instruction pipeline an anti-dependence always creates one or more stalls.
 Which one of above statements is/are correct?
 a) Only S₁ is true b) Only S₂ is true
 c) Only S₁ and S₃ are true d) Only S₂ and S₃ are true
79. Consider the following sequence of Micro-Operations
 MBR ← PC
 MAR ← X
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 Which one of the following is a possible operation performed by this sequence?
 a) Instruction fetch b) Operand fetch
 c) Conditional fetch d) Initiation of interrupt service
80. Register renaming is done in pipelined processor
 a) as an alternative to register allocation at compile time
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 c) to handle certain kinds of hazards
 d) as part of address translation
81. If a transmission has packet delays 10ms, 12ms and 9ms in a communication, then what this indicates in data transmission?
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82. A noisy channel has transmission speed of 2kbps and a bandwidth of 4kHz. What is the channel Signal to Noise Ratio (SNR)
 a) $1 + \sqrt{2}$ b) $\sqrt{2} - 1$ c) $1 - \sqrt{2}$ d) None of these
83. A network with CSMA/CD has a bandwidth of 10Mbps. If the maximum propagation time is 40μsec. What is the maximum frame size?
 a) 10 bytes b) 101 bytes c) 100 bytes d) 1 byte
84. Identify the class of IP-address 193.131.80.255.
 a) Class C b) Class D c) Class A d) Class E
85. An attacker has captured the login credentials of a Banker by posting fake web page, similar to the Bank web page. This type of attack is known as:
 a) Denial of Service (DoS) b) Packet Mistreat
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88. High level synchronization construct to solve critical section problem is called.
 a) Paging b) Swapping c) Monitors d) Testandset
89. Cooperative concurrent reader and writer processes writing to the standard output file leads to.
 a) Race condition b) Output varies on the order of execution of processes
 c) Output is unpredictable d) All of these
90. Consider the virtual page reference string 1, 2, 3, 2, 4, 1, 3, 2, 4, 1 on a demand paged virtual memory system running on a computer system that has main memory size of 3 page frames which are initially empty. Let LRU, FIFO and OPTIMAL denote the number of page faults under the corresponding page replacement policy. Then,
 a) $OPTIMAL < LRU < FIFO$ b) $OPTIMAL < FIFO < LRU$
 c) $OPTIMAL = LRU = FIFO$ d) $OPTIMAL = FIFO$
91. If $A \cap B = A$ then $A \cup B$ is equal to?
 a) A b) ϕ c) A' d) B
92. Which of the following statements is a proposition?
 a) Next month I will pass ETR examination b) Oh, It is awesome
 c) The sum of $x + 5$ is 7 d) $15 \times 321 = 54321$
93. The range of the relation $R = \{(x, x^2) \mid x \text{ is a prime number less than } 13\}$ is
 a) $\{4, 9, 25, 49, 121\}$ b) $\{1, 2, 3, 5, 7\}$
 c) $\{2, 3, 5, 7, 11\}$ d) $\{1, 4, 9, 25, 49, 121\}$
94. In how many ways can 3 or more persons can be selected from 12 persons?
 a) 2200 b) 220×12 c) 12×3 d) 4017

95. Let G be a simple graph of order η . If the size of G is 56 and size of $\overline{G}$ is 80 then what is η ?
a) 34 b) 17 c) $50 - (80/2) = 10$ d) $80 - 56 = 24$
96. _____ is the process that groups data and procedures into an entity called objects.
a) Object development methodology b) Linear programming
c) Structured programming d) Object oriented system development
97. In the _____ phase the design model is built based on the analysis model.
a) System Design b) Application c) Object design d) Analysis
98. _____ deals with the static process view of a system, from the perspective of a real or prototype case.
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99. _____ denotes the aspirations of the users and the responsibility of the system to its users.
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100. The output of object analysis is a description of the _____ and the user requirements.
a) Problem b) Solution c) Quality assurance d) Use cases

VTU-ETR Seat No.

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Question Paper Version : D

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) November – 2022

**Computer Science and Engineering /
Information Science and Engineering**



Time: 3 hrs.

Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
8. Candidates should use black ball point pen to shade the checkbox bubble so that it is fully dark and the computer can detect without fail. Shading should be limited to the checkbox bubble only.
9. Shading of multiple checkbox bubbles shall be treated as invalid answer.
10. Partial or incomplete shading/darkening of the bubble be avoided.
11. Use of whiteners/correction fluid to change the shaded/filled checkbox bubble to fill another is prohibited.
12. Erasing of filled checkbox bubble/s to fill another is prohibited.
13. Any type of writing, scribbling etc., on the question paper and ticking the response of a question/s are prohibited. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
14. Only non-programmable scientific calculator is permitted. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
15. Folding/crushing/mutilating of OMR sheet is not permitted.
16. Handover the OMR sheet to the Room Invigilator while leaving the examination hall.

PART – I
(Research Methodology)

[50 MARKS]

1. What is the function of a post test in ANOVA
 - a) Describe those groups that have reliable difference between group means
 - b) Set the critical value for the 'F' test (or Chi – Square)
 - c) Determine if any statistically significant group differences have occurred
 - d) None of these
2. In ANOVA procedure 'Factor' refers to
 - a) The dependent variable
 - b) The independent variable
 - c) Different levels of a treatment
 - d) The critical value of F
3. In the ANOVA 'treatment' refers to
 - a) Experimental units
 - b) Different levels of a factor
 - c) a factor
 - d) Applying antibiotic to a wound
4. When conducting an ANOVA , F DATA will always fall within what range?
 - a) Between 0 and infinity
 - b) Between 0 and 1
 - c) Between negative infinity and infinity
 - d) Between 1 and infinity
5. In one way ANOVA which of the following is used within F – ratio as a measurement of the variance F individual observations
 - a) SSE
 - b) MSE
 - c) MSRT
 - d) None of the above
6. In regression analysis, the variable that is being predicted is
 - a) The independent variable
 - b) The dependent variable
 - c) Usually denoted by x
 - d) Usually denoted by r
7. If the regression equation is equal to $Y = 23.6 - 54.2x$, then 23.6 is the ____ while – 54.2 is the ____ of the regression line.
 - a) Slope, intercept
 - b) Slope, regression coefficient
 - c) Intercept , slope
 - d) Radius, intercept
8. How many coefficients do you need to estimate in a simple linear regression model (one independent variable).
 - a) 1
 - b) 2
 - c) 3
 - d) 4
9. Which of the following is not the part of the exploratory factor analysis process
 - a) Extracting factors
 - b) Determining the number of factors before the analysis.
 - c) Rotating the factors
 - d) Refining and interpreting the factors
10. If a researcher wanted to determine which variable were associated with which factors she/he would look at
 - a) Factor scores
 - b) Factor loadings
 - c) Factor
 - d) Factor associations
11. Research conducted to solve an immediate problem is
 - a) Fundamental Research
 - b) Analytical Research
 - c) Action Research
 - d) Survey

12. Research pertaining to pure mathematics or natural laws is the example of
 - a) Qualitative Research
 - b) Empirical Research
 - c) Analytical Research
 - d) Fundamental Research
13. Research to study the effect of certain Policies, Plans and Program is
 - a) Applied Research
 - b) Descriptive Research
 - c) Evaluation Research
 - d) Causal Research
14. The first step of Research process is
 - a) Research design
 - b) Analysis of data
 - c) Review of literature
 - d) Identification of research problem
15. Which of the following helps to compare two or more variables
 - a) Classification
 - b) Tabulation
 - c) Survey
 - d) Literature review
16. Choose the best answer, which is the most reliable source of information for your literature review :
 - a) A TV documentary
 - b) A Newspaper article
 - c) A peer reviewed research article
 - d) A relevant chapter from a text book
17. Which literature review involves time line based collection of literature review
 - a) Narrative method
 - b) Systematic method
 - c) Meta – analysis method of literature review
 - d) Meta – synthesis method of literature review.
18. Narrative literature review method is also known as
 - a) Advanced method
 - b) Scientific method
 - c) Traditional method
 - d) Systematic method.
19. A review of the literature should sort out the information into a _____ framework
 - a) Conceptual
 - b) Selective
 - c) Theoretical
 - d) Universal
20. The internet should not be used for finding published literature
 - a) True
 - b) False
21. What is a complementary event for “At least one head appears” if two coins are tossed simultaneously?
 - a) Exactly one head appears
 - b) Atleast one tail appears
 - c) Atmost one tail appears
 - d) None head appears
22. A bag contains 7 Red and 4 Blue Balls. Two Balls are drawn at random with replacement. The probability of getting the Balls of different colors is :
 - a) $28/121$
 - b) $56/121$
 - c) $\frac{1}{2}$
 - d) None of the above
23. If two coins are tossed simultaneously, what is the probability of getting atleast one head?
 - a) $\frac{1}{2}$
 - b) $\frac{1}{3}$
 - c) $\frac{1}{4}$
 - d) $\frac{3}{4}$
24. What are Independent Events?
 - a) If the outcome of one event does not affect the outcome of another
 - b) If the outcome of one event affects the outcome of another
 - c) Any one of the outcome of one event does not affect the outcome of another.
 - d) Any one of the outcome of one event does affect the outcome of another

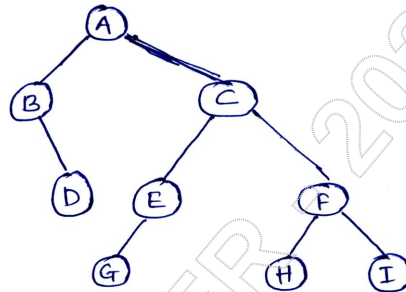
25. In class, 30% of students study Hindi, 45% study Maths and 15% study Both Hindi and Maths. If a student is randomly selected, what is the probability that he/she study Hindi and Maths?
 a) $1/5$ b) $3/5$ c) $2/5$ d) $2/7$
26. The normal distribution
 a) Mean = Variance = 1 b) Standard deviation = Variance = 0
 c) Mean = 0, Variance = 1 d) Mean = Standard deviation
27. The shape of normal distribution curve
 a) Bell shaped b) Flat c) Circular d) Spilled
28. In a Binomial distribution, if 'n' is the number of trials and 'p' is the probability success, then the mean value is
 a) np b) n c) p d) np (1-p)
29. If 'm' is the Poisson distribution, the standard deviation is
 a) $(\sqrt{m})$ b) m^2 c) $\bar{m}$ d) m/2
30. The mean of hypergeometric distribution is
 a) $n * K / N - 1$ b) $n * K - 1 / N$ c) $n - 1 * K / N$ d) $n * K / N$
31. Which among the following is a probability sampling method?
 a) Judgemental sampling b) Quota sampling
 c) Simple random sampling d) Convenience sampling
32. The difference between a static and the parameters is called as
 a) Non - random b) Probability c) Sampling error d) Random
33. Sampling error increases as we increases the sampling size
 a) True b) False c) True to some extent d) None of the above
34. Population under research study is referred as
 a) Standard Population b) Final Population
 c) Infinite Population d) Target Population
35. The process of finding true but unknown value of population parameter is called as
 a) Statistical inference b) Estimation
 c) Guess d) Biased
36. Arrange the following steps in the process of hypothesis testing in proper sequence
 A) Select the level of significance B) Setup null and alternative hypothesis
 C) Establish the decision rule D) Performance computations
 E) Select test statistics F) Draw conclusion
 Choose the correct answer from the options given below :
 a) A, B, C, D, E, F b) A, B, E, D, C, F c) B, A, C, D, E, F d) B, A, E, C, D, F
37. Which among the following is non parametric statistics?
 a) t - test b) F - test
 c) 2 - test d) Spearman's rank order correlation
38. Which test is normally applied for qualitative data :
 a) 't' test b) 'F' test c) X^2 Chi – square test d) 'z' test

39. An investigator commits Type – 1 error in testing hypothesis when he / she
a) Accepts null hypothesis when it is false b) Rejects null hypothesis when it is true
c) Accepts null hypothesis when it is true d) Rejects null hypothesis when it is false
40. If the critical region is located equally in both sides of the sampling distribution of test statistic, the test is called
a) One tailed b) Two tailed c) Right tailed d) Left tailed
41. Which one among the following is the most comprehensive source of population data?
a) National sample surveys b) Demographic health surveys
c) National family health surveys d) Census
42. Which one is called non – probability sampling?
a) Cluster sampling b) Systematic sampling
c) Stratified sampling d) Quota sampling
43. Which scale is the simplest form of measurement
a) Interval b) Ratio c) Nominal d) Ordinal
44. Absolute zero exists in
a) Ratio scale b) Interval scale c) Nominal scale d) Ordinal scale
45. Which data is cheaper to collect?
a) Primary data b) Collective data c) New data d) Secondary data
46. After the data has been processed and analysed, the research process requires
a) Interpretation of data b) Presentation of data
c) Reporting of data d) Testing of data
47. The process of marking segments of data with symbols, descriptive words or category names is known as
a) Concurring b) Coding c) Coloring d) Segmentation
48. The most repeated value in the series is called as
a) Mean b) Mode c) Median d) Range
49. A study in statistics that helps to interpret the variability of data known as
a) Standard deviation b) The measure of central tendency
c) The measure of dispersion d) None of the above
50. For a frequency distribution of a variable X, mean = 32 and median = 30. The distribution is
a) Positive skewed b) Negatively skewed c) Mesokurtic d) Platykurtic

PART – II
Discipline Oriented Section
(Computer Science and Engineering /
Information Science and Engineering)

[50 MARKS]

51. What is the post order traversal of following binary tree?

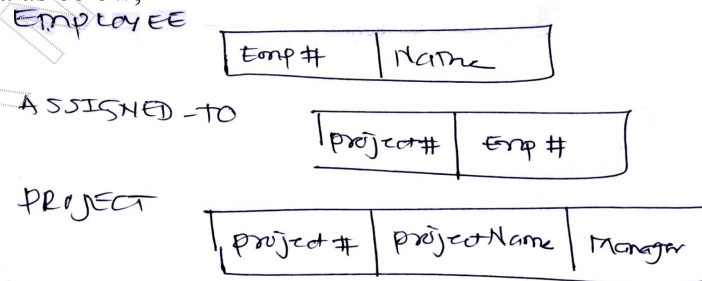


- a) A B D C G E H I F b) D B G E H I F C A
c) D B G E F H I C A d) A B C D E F G H I
52. Insertion of an element at the middle of a singly linked list requires the modifications of how many pointers?
a) 2 b) 1 c) 3 d) 4
53. Which type of traversal of binary search tree outputs the value in sorted order?
a) Pre order b) In order c) Post order d) None
54. The searching operation in an array is done using.
a) Key and Index b) Only key c) Only Index d) None of these
55. A binary search tree is generated by inserting in order the following integers (left to right). 50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24. The number of nodes in the left subtree and right subtree of the root respectively is.
a) (4, 7) b) (8, 3) c) (7, 4) d) (3, 8)
56. Consider the following processor design characteristics:
I. Register – to – Register arithmetic operation only.
II. Fixed-length instruction format.
III. Hardwired control unit.
Which of the characteristics above are used in the design of a RISC processor?
a) I and II only b) II and III only c) I and III only d) I, II and III
57. A processor has 40 distinct instruction and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is _____.
a) 16 bits b) 17-bits c) 18-bits d) 19-bits

58. Consider the following code sequence having five instructions I₁ to I₅, each of these instruction has the following format:
 OP R_i, R_j, R_k
 Where operation OP is performed on contents of register R_j and R_k and the result is stored in register R_i.
 I₁ : ADD R₁, R₂, R₃
 I₂ : MUL R₇, R₁, R₃
 I₃ : SUB R₄, R₁, R₅
 I₄ : ADD R₃, R₂, R₄
 I₅ : MUL R₇, R₈, R₉
 Consider the following three statements:
 S₁ : There is an anti-dependence between instruction I₂ and I₅
 S₂ : There is an anti-dependence between instruction I₂ and I₄
 S₃ : Within an instruction pipeline an anti-dependence always creates one or more stalls.
 Which one of above statements is/are correct?
 a) Only S₁ is true
 b) Only S₂ is true
 c) Only S₁ and S₃ are true
 d) Only S₂ and S₃ are true
59. Consider the following sequence of Micro-Operations
 MBR ← PC
 MAR ← X
 PC ← Y
 Memory ← MBR
 Which one of the following is a possible operation performed by this sequence?
 a) Instruction fetch
 b) Operand fetch
 c) Conditional fetch
 d) Initiation of interrupt service
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 a) as an alternative to register allocation at compile time
 b) for efficient access to function parameter and local variable
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 d) as part of address translation
61. If $A \cap B = A$ then $A \cup B$ is equal to?
 a) A
 b) ϕ
 c) A'
 d) B
62. Which of the following statements is a proposition?
 a) Next month I will pass ETR examination
 b) Oh, It is awesome
 c) The sum of x + 5 is 7
 d) $15 \times 321 = 54321$
63. The range of the relation $R = \{(x, x^2) \mid x \text{ is a prime number less than } 13\}$ is
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65. Let G be a simple graph of order η . If the size of G is 56 and size of $\overline{G}$ is 80 then what is η ?
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66. _____ is the process that groups data and procedures into an entity called objects.
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70. The output of object analysis is a description of the _____ and the user requirements.
 a) Problem b) Solution c) Quality assurance d) Use cases
71. The relational operator <OP> described below represents the following:
 $R = P \text{ <OP> } Q$
 $R = \{t \mid t \in p \wedge t \notin \theta\}$ and
 $0 \leq |R| \leq |P|$
 a) The difference '____' operator b) The projection 'π' operator
 c) The division '÷' operator d) None of the above

72. Consider the schema as below,



The query

EMPLOYEE $\bowtie$ (ASSIGNED - TO $\div \pi_{\text{project\#}}(X)$)

$X = \sigma_{(\text{project\#} = 'P_1' \vee \text{project\#} = 'P_2')}$ (ASSIGNED-TO) will results in to:

- a) Gathering details at all employees.
 b) Gathering details at all employees working on project 'P₁' or project 'P₂'.
 c) Gathering details at all employees working on both project 'P₁' and 'P₂'.
 d) None of these.
73. Consider the following schema employees:
- | | | | |
|------|------|-----|-----|
| Emp# | Name | DOB | Sex |
|------|------|-----|-----|
- The SQL provision to ensure the attribute 'sex' takes the values either 'M' or 'F' is
 a) ensure (sex in ('M', 'F')) b) insist (sex in ('M', 'F'))
 c) check (sex in ('M', 'F')) d) Need to write stored procedure.

74. Stored procedures are used to:
- enforce business rules and improve performance
 - to make code easier to maintain.
 - code will be stored once for use by multiple applications
 - All of these
75. To set transaction in read only access mode is through:
- Setting isolation level in read uncommitted
 - Setting isolation level in read committed
 - Setting isolation level in serializable
 - Setting isolation level in repeatable read mode.
76. Customers arrive at a reception counter at an average interval of 10 minutes and the receptionist takes an average of 6 minutes for one customer. Determine the average queue length.
- 9/10
 - 7/10
 - 11/10
 - 3/10
77. Which of the following statements are not true of simulation?
- Simulation model cannot prescribe what should be done about a problem.
 - Simulation models can be used to study alternative solutions to a problem.
 - Simulation models the behavior of a system.
 - The equations describing the operating characteristics of the system are known.
78. Simulation models can be used to obtain operating characteristics in less time than with real system using a feature of simulation called.
- Microseconds
 - Warp seed
 - Time compression
 - None of the above
79. A multiple-choice test has 20 questions. There are four choices for each question. A student who has not studied for the test decides to answer all the questions randomly by guessing the answer for each question. Which of the following probability distributions can be used to calculate the students chance of getting at least 20 questions right?
- Binomial Distribution
 - Poisson Distribution
 - Exponential Distribution
 - Normal Distribution
80. Which of the following are techniques to perform verification of simulation model?
- By using programming skills to write and debug the program in sub-programs.
 - By tracing the intermediate results and comparing them with observed outcomes.
 - By checking the simulation model output using various input combinations.
 - All of these.
81. What is the minimum time required to compute power (x, n) is x^n in the options given below?
- $O(\log n)$
 - $O(n)$
 - $O(\log \log n)$
 - $O(n \log n)$
82. n strings each of length n are given. What is time taken to sort the strings in lexicographic order using 2 way merging?
- $O(n \log n)$
 - $O(n^2 \log n)$
 - $O(n \log \log n)$
 - $O(n \log^2 n)$
83. Number of spanning tree for complete graph with six nodes is
- 1296
 - 1398
 - 1354
 - 1366

84. Number of ways of parenthesizing 6 matrices is
a) 36 b) 42 c) 64 d) 78

85. For a job assignment problem

J_1	J_2	J_3	J_4	
10	2	7	8	P_1
6	4	3	7	P_2
5	8	1	8	P_3
7	6	10	4	P_4

Total cost of optimal (least cost) assignment is

- a) 10 b) 18 c) 13 d) 14
86. In software engineering the most popular term COTS stands for
a) Citation On The System b) Commercial On The System
c) Commercial Off The Shelf System d) Central Operating Trading System

87. Form based requirement specification in software engineering is classified under
a) New Language Specification b) Structured Natural Language
c) Graphical Notations d) Mathematical Specifications

88. Sequence diagrams used in software development are classified under
a) Static model b) Dynamic model
c) Probabilistic model d) Stochastic model

89. In the following software testing method individual program units or object classes are tested
a) Unit testing b) Component testing c) System testing d) All of these

90. Software pricing depends upon
a) Travel cost b) Hardware and software cost
c) Training cost d) All of these

91. If a transmission has packet delays 10ms, 12ms and 9ms in a communication, then what this indicates in data transmission?
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a) $1 + \sqrt{2}$ b) $\sqrt{2} - 1$ c) $1 - \sqrt{2}$ d) None of these

93. A network with CSMA/CD has a bandwidth of 10Mbps. If the maximum propagation time is 40μsec. What is the maximum frame size?
a) 10 bytes b) 101 bytes c) 100 bytes d) 1 byte

94. Identify the class of IP-address 193.131.80.255.
a) Class C b) Class D c) Class A d) Class E

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a) OPTIMAL < LRU < FIFO b) OPTIMAL < FIFO < LRU
c) OPTIML = LRU = FIFO d) OPTIMAL = FIFO